Cut Free Pets — COMPLETE PROJECT REQUIREMENTS & GAME BIBLE

Working title: Cut Free Pets (final store name later)
Document type: Full requirements + design + tech + data + auth + privacy + build bible
Audience: Solo developer
Version: 4.4 — Asset file catalog + Script→Function map + 2D lock + Steps 0–28 Play Store ready
Rule: This file is the single source of truth. If it is not here, it is not in scope until added.

Important locked removals

System	Status
Stars / star unlock / best stars	**REMOVED**
Attempts / lives / continue-for-lives	**REMOVED**
Dog unlock	Clear **Level 10** (tunable), not stars
Scary body-tied cat / messy knot ball	**REJECTED** (bad UX / hard to read)
Static closed crate with no movement	**REJECTED** (boring)

Locked core fantasy (v4.0) — READ THIS FIRST

A cute cat sits in an OPEN hanging box/basket so the full cat is always easy to see.
Numbered ropes hold the box up.
Player cuts ropes in order (1 → 2 → … → final).
Each non-final cut lowers the box one step.
The FINAL rope cut drops the box fully down → cat becomes free and happy.

Feel goal (like Mob Control ads):

- Every cut = mini dopamine (snip + drop + cat reaction)
- Final cut = big payoff (full drop + free celebration)
- First levels easy; understand in 1 second

Visual / animation lock: Same approach as **Cut the Rope** — **2D sprites + 2D animation** (not 3D models). Flat cute characters, 2D ropes, DOTween motion, sprite Animator / flipbook reactions.

Table of Contents

A. Product

1. Game Identity
2. Goals & Success Criteria
3. Locked Decisions
4. In Scope / Out of Scope
5. Functional Requirements (Master List)
6. Non-Functional Requirements

B. Game Design

7. How The Game Works (Hanging Box)
   7A. Gaming Screen Layout
   7B. Mob Control–Style Feel Rules
8. Cut System + Box Drop Steps
9. Rope & Chain System
10. Animal + Box Behavior
11. Progress & Dog Unlock (No Stars)
12. Fail / Oops / Restart (No Attempts)
13. Hint System
14. Ads System

C. User, Account & Data

15. Player Modes (Guest vs Google)
16. Google Sign-In Requirements
17. User Data Model (Local + Cloud)
18. Sync / Merge Rules
19. Privacy, Consent & Legal
20. Analytics & Events
21. Permissions

D. UX / Screens

22. Full User Journeys
23. Screen Map & Navigation

24. Screen Specs (Including Account)

E. Content

25. Level Design Spec

F. Technical

26. Tech Stack
27. Backend / Services
28. Project Folder Structure
29. Code Architecture
30. Scripts Responsibility
31. Level Data Format
32. Scenes & Prefabs
33. Animation / Audio / Art
34. Camera, Input, UI, Performance Rules

G. Delivery

35. Build Order (High Level)
    35A. Detailed Implementation Build Steps (Step 0–28 → Play Store Ready)
36. Testing & Acceptance
37. Play Store Requirements
38. Marketing Ads Spec
39. Risks & Edge Cases
40. Later Roadmap
41. Glossary
42. One-Page Summary
43. Still Pending (Does Not Block Coding)
44. Asset File Catalog (Exact Names)
45. Script → Function Map (Which Function in Which File)

A. PRODUCT

1. Game Identity

1.1 One sentence

Modern casual puzzle: swipe to cut numbered ropes holding a **hanging open box** with a fully visible cute cat; each cut lowers the box; the **final cut** drops it and frees the cat.

1.2 Genre / feel

Casual puzzle with dopamine of **Mob Control / Amaze GO** ads (instant understand + satisfying moments), rope snip feel of **Cut the Rope**.
Fun = snip + visible box drop + cute free moment, **not** points.

1.3 Platform

Item	Requirement
Primary	Android (Google Play)
Secondary later	iOS
Orientation	Portrait
Language	English UI only (v1)
Offline	Core gameplay works offline; login/ads/cloud need network

1.4 Monetization

Free + ads (Hint rewarded + interstitial every **4** wins after first ~8 protected).
No paywall to play core levels in v1.

---

2. Goals & Success Criteria

2.1 Product goals (priority order)

1. Player enjoys snip + free moment
2. Understands in < 3 seconds
3. Wants “one more level”
4. Downloads grow
5. Fair ads revenue

2.2 Acceptance “fun gate”

Ask 5 testers after 2 minutes:

- Understood without help?
- Snip feels good?
- Cute/fun?
- Want next?

Need mostly yes before mass levels/ads polish.

2.3 Soft launch quality gates

- Crash-free session rate high
- No star/attempt UI anywhere
- Progress saves correctly guest + Google
- Ads do not show on Oops/Restart

---

3. Locked Decisions

Topic	Decision
Core scene	**Open hanging box/basket + fully visible cat**
Trap logic	Ropes hold box up (not scary body-tying)
Cut order	Numbered ropes; only next rope cuttable
Per cut	Box **lowers one step** + snip juice + cat reacts
Final cut	Box **fully drops/lands** → cat free/happy
Visuals	**2D only** — Cut the Rope style (sprites + 2D anim, not 3D)
Animation	**2D** sprite Animator / flipbooks + DOTween (rope snip, box drop, cat react)
Rope feel	Cut-the-Rope snip/split/fall (2D LineRenderer or 2D segments)
Cut input	Swipe + scissors ~0.2s
Animals	Cat → Dog → third later
Levels v1	20–30
Stars	None
Attempts	None
Dog unlock	Clear Level 10
Wrong rope Levels 1–5	**Ignore** (no fail)
Wrong rope Levels 6+	Soft fail: box fail wobble + sad cat + reset (no ad)

Box motion	Drop steps down; prefer small tilt→straighten OR rails upright
Account	**Guest play + optional Google Sign-In**
Cloud save	Yes when Google linked (see §15–18)
Budget start	$0 assets; Play Console $25 when publishing
Name	Later

4. In Scope / Out of Scope

4.1 In scope (v1 / v1.1)

- Hanging open box gameplay with drop steps
- Numbered ordered rope cuts
- Final-rope full drop + free celebration
- 20–30 levels, ropes + chains
- Guest local progress
- Optional Google login + cloud progress sync
- Hint ads + win interstitials
- Settings, privacy links, basic analytics
- Android release

4.2 Out of scope (v1)

- Stars / lives systems
- Forced login before play
- Facebook/Apple login (Apple later with iOS)
- Multiplayer / chat / friends list
- Spinner
- Zoo decorate meta
- Realistic hanging-weight physics debates (use clear step drops)
- **3D character models / 3D main look** (locked to Cut the Rope–style **2D**)
- iOS (until Android stable)
- Real-money gambling mechanics

5. Functional Requirements (Master List)

FR-GAME

ID	Requirement
FR-G01	Player can swipe to cut a rope/chain
FR-G02	Scissors FX ~0.2s plays on valid cut (**2D** sprite anim)
FR-G03	Rope splits and falls (not instant delete) — **2D** pieces
FR-G03b	Game visuals are **2D Cut the Rope style** (sprites + 2D Animator; not 3D)
FR-G04	Cat is in an **open hanging box**; full cat always readable
FR-G05	Ropes are numbered; only the next ordered rope is cuttable
FR-G06	Each valid **non-final** cut lowers the box one visible step
FR-G07	**Final** rope cut fully drops/lands the box and frees the cat
FR-G08	Wrong-order cut: Levels **1–5 ignore**; Levels **6+** Oops + reset (unlimited, no ad)
FR-G09	Unlimited retries (no attempts)
FR-G10	Restart reloads level without ad
FR-G11	Hint via rewarded ad highlights next rope
FR-G12	Win shows celebration + Next (no stars)
FR-G13	Level 1 shows swipe tutorial hand once
FR-G14	Cat chapter starts available
FR-G15	Clearing Level 10 unlocks Dog chapter
FR-G16	Chains appear in later levels with clank SFX
FR-G17	Gameplay works offline
FR-G18	Each cut plays snip juice + cat reaction (mini-win feel)

FR-PROGRESS

ID	Requirement
FR-P01	Level 1 unlocked by default
FR-P02	Clearing level K unlocks K+1
FR-P03	Cleared levels show checkmark (not stars)

FR-P04	Progress saved on device for guest
FR-P05	Progress syncs to cloud when Google signed in
FR-P06	Replaying cleared levels always allowed free

FR-ADS

ID	Requirement
FR-A01	No interstitial on first ~8 wins
FR-A02	Interstitial every 3–4 wins after protection
FR-A03	Never interstitial on Oops/Restart/fail
FR-A04	Hint uses rewarded ad
FR-A05	No banner during gameplay
FR-A06	Test ad IDs in debug; real IDs in release

FR-ACCOUNT

ID	Requirement
FR-U01	User can play as Guest without login
FR-U02	User can Sign in with Google from Home/Settings
FR-U03	After Google login, progress can sync to cloud
FR-U04	User can Sign out (guest local remains policy below)
FR-U05	User can request account/data delete (Play policy)
FR-U06	Login is never required to start Level 1
FR-U07	Show signed-in Google account name/email (masked ok)

FR-SETTINGS

ID	Requirement
FR-S01	Music on/off
FR-S02	Sound on/off
FR-S03	Haptics on/off optional
FR-S04	Open Privacy Policy
FR-S05	Open Terms (if used)

FR-S06	Account section: Sign in / Sign out / Delete

6. Non-Functional Requirements

ID	Area	Requirement
NFR-01	Performance	Stable play on mid Android; avoid heavy lag
NFR-02	Startup	Reach Home quickly; Splash ≤ ~2s perceived
NFR-03	Reliability	Progress must not randomly reset
NFR-04	Security	Do not store Google password; use OAuth tokens via SDK
NFR-05	Privacy	Collect minimum data; disclose in policy + Data Safety
NFR-06	Accessibility	Large buttons; readable ropes; English clear labels
NFR-07	Localization	English only v1; structure text for later
NFR-08	Maintainability	Clear folders/scripts; no god-class
NFR-09	Network	Graceful offline: gameplay OK; login/sync/ads fail softly
NFR-10	Versioning	save_version field for migrations

B. GAME DESIGN

7. How The Game Works (Hanging Box)

7.1 Core loop

```
Load level
→ Open hanging box high up + fully visible cat + numbered ropes
→ Player swipes next rope (only #current is active)
→ Valid cut: scissors + snip SFX + rope falls
→ If NOT final rope: box lowers one step + cat reacts (mini-win)
→ If FINAL rope: box fully drops/lands + cat free/happy + WIN UI
→ Invalid order: Levels 1–5 ignore; Levels 6+ Oops + reset (unlimited, no ad)
→ Next level
```

7.2 Win condition

Final required rope is cut → box completed its drop sequence → cat freed.

7.3 Why final rope controls the full drop

In real physics, cutting early ropes might drop everything too soon.
Game rule (player-readable):

- N ropes = N drop steps
- Cuts 1..(N-1) = partial lower only
- Cut N (final) = full drop + free

Optional visual aid: wall height marks (High / Mid / Low / Free).

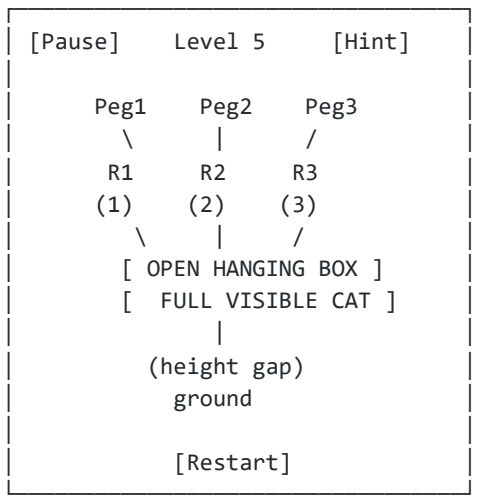
7.4 Level length target

15–40 seconds.

7.5 Chapters

- Cat from start
- Dog after Level 10 cleared (same hanging-box fantasy, dog inside)
- Third animal later

7A. Gaming Screen Layout



Screen rules

Element	Rule
Cat	Always fully visible in open box/basket (not sealed dark crate)

Box	Cute open front / basket so face+body readable
Ropes	Thick, numbered, only next one bright
Center	Keep clear — no big panels over cat/box
HUD	Pause, Level, Hint, Restart only
Forbidden	Stars, hearts, lives, clutter

Height progression (must be obvious)

After cuts	Box height
0 cuts	High
Each mid cut	One step lower
Final cut	Landed / fully down

7B. Mob Control–Style Feel Rules (Max Users)

People like Mob Control because of **instant visible dopamine**, not scoreboards. Map that here:

Mob Control feeling	Our game
Something changes every action	Box drops each cut
Power rising	Height lowering toward free
Smash payoff	Final drop + cat free
Instant understand	See hanging box + ropes
Fast first win	Level 1 free in < 15–20s

Must juice every cut

1. Scissors ~0.2s
2. Strong snip SFX (variants)
3. Rope split/fall
4. Box tween down one step
5. Tiny camera punch
6. Cat reaction (meow / ears / paw)

Must juice final cut (bigger)

1. Stronger snip
2. Bigger camera punch
3. Box drops with weight + dust on land
4. Cat jumps out happy + hearts
5. “Freed!” → Next

First-session rules for likes/downloads

- Levels 1–5 very easy
- Ghost hand on Level 1
- No interstitial early
- No fail walls
- Ad creatives = real snip → drop → free montage

---

8. Cut System + Box Drop Steps

8.1 Valid cut timeline

Step	Time	What
Swipe trail	continuous	Show trail
Hit active rope	0	Validate order index
Scissors	0.00–0.20	Snip at point
Rope split/fall	0.10–0.70	Pieces fall
Box drop step	0.15–0.55	Lower to next height (if not final)
Cat react	overlap	IdleHope / meow
If final	0.20–0.90	Full drop + land FX
If final	+0.1	Happy free + Win UI

8.2 Order rules

- `currentOrder = 1` at level start
- Only rope with `orderIndex == currentOrder` is cuttable (bright)
- Others dim
- On success: `currentOrder++`
- When last rope cut: trigger final drop + win

8.3 Wrong rope rules (LOCKED)

Levels	Wrong rope swipe	Result
1–5	**Ignore**	No cut, no fail, no reset
6+	Soft fail	Box fail wobble + cat sad + Oops + auto reset ropes/box/order

Constant: `WrongRopeFailFromLevel = 6`
Never show ad on wrong-rope fail. No attempts.

8.4 Tiny input rules

- Min swipe distance
- Block new cuts while scissors/box tween plays
- Disable cuts during win

---

9. Rope & Chain System

Must

- Soft hang / slight sway
- Cut at swipe point
- Halves fall
- Sound synced
- Number badge 1..N

Materials

Type	When	Sound
Rope	Early–mid	Soft snip
Chain	Later / often final lock	Clank

Implementation stages

- **A:** LineRenderer/segments + DOTween rope fall + DOTween box Y steps
- **B:** Verlet/physics rope upgrade later

---

10. Animal + Box Behavior

Box states

State	Meaning
HangHigh	Level start
HangStepK	After K non-final cuts
Dropping	Final cut animation
Landed	On ground / free zone
OpenFree	Cat exits (optional lid/side open)

Cat states

State	When
IdleHang	Waiting in box
HopeReact	After each mid cut
ExcitedLow	Box near ground
HappyFree	After final land
OopsReact	Wrong cut (optional)

2D only: Unity Animator on sprite sheets / flipbooks (Idle / Hope / Sad / Happy).
DOTween for box Y steps, scissors, UI — not 3D skeletal/mesh.
Cat must remain fully visible at all times until free exit celebration.

---

11. Progress & Dog Unlock (No Stars)

Track:

- `level_cleared[id]`
- highest unlock
- `dog_unlocked`

Dog:

```
if level 10 cleared → dog_unlocked = true
```

UI: checkmarks, not stars.
Win UI: no star row.

---

12. Fail / Oops / Restart (No Attempts)

No hearts, no attempt counter, no continue-lives ad.
Restart always free, restores full level state.

Wrong-rope fail flow (levels 6+ only)

1. Player swipes inactive/wrong rope
2. Box small shake / fail wobble
3. Cat sad (0.5–1.0s)
4. “Oops!”
5. Auto reset: ropes restored, box to start height, `currentOrder = 1`
6. Player continues immediately
7. **No ad**

Levels 1–5: wrong rope ignored (optional tiny arrow toward active rope).

---

13. Hint System

Hint → rewarded ad → glow correct rope 3–5s.
No auto-cut. Soft fail if ad not ready.

---

14. Ads System

Ad	Rule
Rewarded Hint	Optional
Interstitial	Every 3–4 wins after early protection
On Oops/Restart	Never
Banner gameplay	No

Use AdMob. Never click own ads.

C. USER, ACCOUNT & DATA

15. Player Modes (Guest vs Google)

15.1 Guest (default)

- No login required
- Progress stored **locally on device**
- Can play full game
- Risk: uninstall/new phone may lose progress

15.2 Google signed-in

- Optional
- Progress stored **local + cloud**
- Restore on new device after login
- Better long-term retention

15.3 Recommended UX policy (locked)

Never force Google login before first play.
Show soft prompt after Level 5 or on Home:
"Sign in with Google to keep your progress safe."
Dismissible.

---

16. Google Sign-In Requirements

16.1 Provider

- **Google Sign-In** on Android
- Recommended stack for solo: **Firebase Authentication (Google provider)**
- Alternative: Google Play Games Services login (optional later; different UX)

16.2 What Google gives the app

Data	Use
Google user id (uid / sub)	Primary account key
Email	Account display / support (optional show masked)
Display name	Optional greet

Profile photo URL	Optional avatar on Home
OAuth tokens	Handled by Firebase/SDK (do not custom-store passwords)

16.3 What we never do

- Never ask for Google password in our UI
- Never sell personal data
- Never require login to cut ropes

16.4 Auth flows

Sign in

```
User taps Sign in with Google
→ Google account picker
→ Success → Firebase Auth session
→ Create/load cloud profile
→ Merge with local progress (see §18)
→ UI shows signed-in state
```

Sign out

```
User taps Sign out
→ Clear auth session
→ Keep local progress copy on device
→ Cloud remains for next login
→ UI returns to Guest mode label
```

Fail cases

Case	App behavior
No network	Toast: “Connect to internet to sign in”
User cancels picker	No change
Auth error	Soft error message; stay Guest
Play Services missing	Soft message / open update

16.5 Account delete (required for stores when account exists)

```
Settings → Delete account
→ Confirm dialog (clear warning)
→ Delete cloud user data
→ Delete/sign out auth user
→ Reset or keep local guest? → Policy: reset cloud; offer keep local cleared copy as guest OR wip
```

Locked default delete policy:

1. Delete cloud profile + auth
2. Sign out
3. Keep local device progress as Guest unless user checks “Also delete local progress”

---

17. User Data Model (Local + Cloud)

17.1 Local device save (always)

Technology: PlayerPrefs **or** JSON file in `persistentDataPath` (JSON preferred once cloud exists for easier merge).

LocalProfile

```
save_version: int                  // current = 4
install_id: string                 // random UUID generated first launch
auth_uid: string | empty           // Google/Firebase uid if linked
is_guest: bool
display_name_cache: string
avatar_url_cache: string
email_cache: string                // optional; consider hashing/masking in logs

progress:
  highest_unlocked_level: int
  dog_unlocked: bool
  third_animal_unlocked: bool      // future
  levels: map<levelId, LevelProgressLocal>
    LevelProgressLocal:
      cleared: bool
      cleared_at_utc: string optional
      play_count: int optional
      last_played_at_utc: string optional

settings:
  music_on: bool
  sound_on: bool
  haptics_on: bool
  consented_privacy: bool
  consented_privacy_version: string
  analytics_opt_in: bool           // if you offer toggle where required

ads:
  lifetime_wins: int
  wins_since_interstitial: int

meta:
  first_open_utc: string
  last_open_utc: string
  app_version: string
  tutorial_level1_done: bool
```

17.2 Explicitly NOT stored

- Stars / mistakeCount for scoring
- Attempts/lives
- Google password
- Full payment card data
- Precise mid-rope cut state for resume (v1 restarts level)

17.3 Cloud profile (when Google signed in)

Recommended: **Cloud Firestore** document
Path: `users/{auth_uid}`

CloudUserDoc

```
uid: string
email: string | null
display_name: string | null
photo_url: string | null
created_at: timestamp
updated_at: timestamp
last_device: string
app_version: string

progress:
  highest_unlocked_level: int
  dog_unlocked: bool
  third_animal_unlocked: bool
  levels: map or array of { id, cleared, cleared_at }
  // store only cleared ids to keep small if preferred

settings_sync_optional:
  music_on, sound_on, haptics_on   // optional; settings can stay device-only

schema_version: int
```

17.4 Server rules (Firestore)

- User can read/write **only** `users/{their_uid}`
- No public progress listing in v1
- Validate types lightly in rules

17.5 Ads / attribution data (third parties)

AdMob / attribution may collect device advertising id, IP, device info per their policies.
Disclose in Privacy Policy + Play Data Safety form.
This is not stored in our custom user doc unless analytics enabled.

17.6 Analytics data (if enabled)

Store/send events, not full personal profiles. See §20.

---

18. Sync / Merge Rules

Critical: avoid progress loss when Guest later signs in.

18.1 When to sync

Event	Action
Google sign-in success	Pull cloud → merge → push

Level win while signed in	Update local + push cloud (debounce OK)
App start signed in + network	Pull soft merge
Sign out	Stop cloud writes; keep local
Offline wins while signed in	Queue local; push when online

18.2 Merge policy (locked)

For each progress field, keep the **more advanced** progress:

Field	Merge
`highest_unlocked_level`	`max(local, cloud)`
`dog_unlocked`	`local OR cloud`
`level.cleared`	`true` if either true
settings	Prefer **local device** settings (do not overwrite mute preference from cloud unless you explicitly sync settings)

After merge: write result to local AND cloud.

18.3 Conflict example

- Phone A guest cleared to Level 12
- Same Google account cloud has Level 8
- Sign in on Phone A → result Level 12 + push cloud

18.4 New phone

- Install → Guest empty
- Sign in Google → download cloud → restore

---

19. Privacy, Consent & Legal

19.1 Required documents

Doc	Need
Privacy Policy URL	Yes (Play)
Terms of Service	Recommended
Data Safety form	Yes

Ads disclosure	Yes

19.2 Privacy Policy must mention

- Guest local progress storage
- Optional Google Sign-In data (id, email, name, photo)
- Cloud save storage location (Firebase/Google Cloud)
- AdMob / advertising identifiers
- Analytics if used
- Account deletion method
- Contact email
- Children's policy stance (family-friendly; follow Play Families rules if targeting kids)

19.3 Consent UX

- First launch: short privacy notice + link + Continue
- If using non-essential analytics in regulated regions, respect consent flags
- Ads SDK consent / UMP (User Messaging Platform) as required by AdMob/region

19.4 Settings links

- Privacy Policy
- Delete account
- Sign in / Sign out

---

20. Analytics & Events

Optional but recommended after core fun works.
Tool: Firebase Analytics (fits Google/Firebase stack).

20.1 Events to log (no stars)

Event	Params
`app_open`	
`tutorial_complete`	level=1
`level_start`	level_id
`level_complete`	level_id, duration_sec
`level_restart`	level_id
`oops`	level_id

`hint_requested`	level_id
`hint_completed`	level_id
`dog_unlocked`	
`sign_in_success`	method=google
`sign_in_cancel`	
`sign_out`	
`cloud_sync_success`	
`cloud_sync_fail`	reason
`ad_interstitial_shown`	
`ad_reward_completed`	type=hint

Do not log passwords or full unnecessary PII in event params.

---

21. Permissions

Permission	Why
Internet	Ads, Google login, cloud sync
Network state	Optional soft offline checks
Ad ID / ATT equivalent	Via ads SDK / policy flows
Vibration	Optional haptics

Do not request contacts, SMS, storage indiscriminately.

D. UX / SCREENS

22. Full User Journeys

22.1 New guest

Install → Splash → Privacy continue → Home → Play → Level Select → Level 1 tutorial → win → Next…
Soft Google prompt after Level 5 (dismissible).

22.2 Guest then Google

Play offline progress → Sign in → merge → cloud protected.

22.3 Returning Google user new phone

Install → Sign in → restore → Level Select shows cleared checkmarks.

22.4 Stuck player

Oops unlimited → Hint ad → win → Next.

22.5 Sign out

Settings → Sign out → Guest label → local progress remains.

22.6 Delete account

Settings → Delete → confirm → cloud wiped → signed out.

---

23. Screen Map & Navigation

```
Splash → Home
Home → Level Select | Settings | Sign-in sheet (modal)
Level Select → Gameplay
Gameplay → Win popup | Pause
Win → Next Gameplay | Replay | Home
Settings → Account actions | Privacy | Toggles | Back Home
```

Android Back:

- Gameplay → Pause
- Popups → close
- Level Select → Home
- Home → system exit confirm optional

---

24. Screen Specs (Including Account)

Splash

Logo, short load, init Save + Auth listener + Ads init (non-blocking).

Home

- Play
- Settings
- Optional Google button / profile chip if signed in
- Optional “Save progress” banner if guest

Level Select

- Cat/Dog sections
- Checkmarks for cleared
- Locks for locked
- Dog lock text: Clear Level 10
- No stars

Gameplay (hanging box scene)

- HUD: Level #, Hint, Restart, Pause only
- Center: open hanging box + fully visible cat + numbered ropes
- Active rope bright; others dim
- Each mid cut: box visibly lowers
- Final cut: full drop + free
- No attempts. No stars.

Win

Box landed, cat happy/out, “Freed!”
Next / Replay / Home. No stars.

Settings

Music, Sound, Haptics
Account: Sign in with Google / Sign out / Delete account
Privacy Policy / Terms

Sign-in modal

Short benefit text + Google button + Not now.

Pause

Resume / Restart / Home.

E. CONTENT

25. Level Design Spec

20–30 levels. **Same fantasy every level:** hanging open box + ordered ropes + drop steps + final drop frees animal.

Design knobs (only these)

1. Rope count (2 → 6)
2. Peg positions (top / left / right)
3. Order sequence
4. Crossings / readability tricks
5. Rope vs chain materials
6. Start height / number of drop steps (= rope count)
7. Background theme
8. Animal (cat / dog)

Curve

Range	Content
1–5	2 ropes, huge clarity, free < 20s, teach swipe + drop
6–10	3 ropes, simple order; Level 10 unlocks Dog
11–20	3–4 ropes, crossings / side anchors
21–30	4–5 ropes, chains as late/final locks, harder layouts

Example Level 1 beat (Mob Control–easy first win)

1. See high box + cat + Rope1/Rope2
2. Ghost hand swipes Rope1
3. Snip + box lowers
4. Swipe Rope2 (final)
5. Full drop + cat free + Next

Design first 10 on paper (peg positions + order only).

F. TECHNICAL

26. Tech Stack

Area	Choice
Engine	**Unity LTS 2D URP** (locked — Cut the Rope style)
Language	C#
Rendering	Orthographic camera, SpriteRenderer, Sorting Layers
Animation	Unity Animator (2D sprites / flipbooks) + DOTween motion
Tweens	DOTween
Ads	Google AdMob
Auth	Firebase Auth + Google Sign-In
Cloud DB	Cloud Firestore
Analytics	Firebase Analytics (optional phase)
Crash	Firebase Crashlytics (optional phase)
Local save	JSON persistentDataPath (preferred) + migration from PlayerPrefs if needed
Git	GitHub Desktop
IDE	VS / VS Code

27. Backend / Services

Firebase project setup checklist

- [ ] Create Firebase project
- [ ] Add Android app (package name must match Unity)
- [ ] Download `google-services.json`
- [ ] Enable Authentication → Google provider
- [ ] Create Firestore database
- [ ] Write security rules (uid-only access)
- [ ] Enable Analytics optional

- [ ] SHA-1/SHA-256 in Firebase for Google Sign-In debug+release

Cost note

Firebase free spark tier usually enough early; monitor reads/writes (debounce sync).

28. Project Folder Structure

```
Assets/
  Art/
    Sprites/       # box, cat, dog, pegs, scissors, UI (2D)
    Animations/    # Animator controllers + clips
    Backgrounds/
  Audio/...
  Prefabs/...
  ScriptableObjects/Levels/
  Scripts/
    Core/
    Cut/
    Rope/
    Animal/        # 2D AnimalController + Animator drivers
    Level/
    UI/
    Ads/
    Save/
    Auth/
    Cloud/
    Analytics/     # optional
    Audio/
  Scenes/
  Plugins/
Docs/
  ASSET_LICENSES.txt
  PRIVACY_POLICY.md (draft)
game-plan/
  FINAL-GAME-PLAN.md   # this file
```

29. Code Architecture

```
Bootstrap
 → SaveManager.LoadLocal
 → AuthManager.Init (listen auth state)
 → AdsManager.Init
 → AnalyticsManager.Init optional
 → Home

Sign-in success
 → CloudSaveService.Pull
 → ProgressMerger.Merge
 → SaveManager.WriteLocal
 → CloudSaveService.Push

Level win
 → SaveManager mark cleared
 → if signed in → CloudSaveService.PushDebounced
 → WinUI
 → AdsManager maybe interstitial
```

Managers: Save, Auth, Cloud, Ads, Audio, GameSession.

30. Scripts Responsibility

Existing gameplay scripts

Gameplay scripts include: `SwipeCutInput`, `Rope*`, `HangBoxController` (height steps + final drop), `AnimalController` (IdleHang / HopeReact / HappyFree), `Level*`, `UI*`, `AdsManager`, `AudioManager`, plus Auth/Cloud scripts below.

Auth / data scripts (ADD)

Script	Responsibility
`AuthManager.cs`	Google sign-in/out, auth state
`CloudSaveService.cs`	Pull/push Firestore
`ProgressMerger.cs`	Merge rules
`LocalSaveRepository.cs`	Read/write JSON
`AccountUI.cs`	Sign in/out/delete UI
`PrivacyGateUI.cs`	First-run notice

`AnalyticsManager.cs`	Event logging optional
`NetworkStatus.cs`	Soft offline checks

`SaveManager.cs` updated: no star fields; owns LocalProfile.

---

31. Level Data Format

```
LevelData (ScriptableObject)
- levelId
- chapter: Cat/Dog
- animalType
- environmentId
- boxStartHeight / dropStepHeights[]   // Y positions after each cut
- ropes: List<RopeData>
- tutorialEnabled

RopeData
- ropeId
- type: Rope/Chain
- orderIndex: 1..N
- pointA (peg), pointB (box attach)
```

No star fields.
`dropStepHeights.Length` should match rope count (or N-1 mid steps + final land).

---

32. Scenes & Prefabs

Scenes: Splash, Home, Game.
Prefabs: `PF_OpenHangBox`, `PF_Cat`, `PF_Dog`, `PF_Rope`, `PF_Chain`, `PF_Peg`, `PF_ScissorsFX`, cut/land particles, UI buttons, account modal.

---

33. Animation / Audio / Art

Visual style lock (Cut the Rope = 2D)

This game uses **the same 2D animation approach as Cut the Rope**:

- Flat **sprite** characters (cat/dog), not 3D models
- **Orthographic** camera

- **2D** ropes (LineRenderer in XY or 2D segment sprites)
- **Sprite Animator** / flipbook for idle + reactions
- **DOTween** for snappy motion (box drop steps, scissors, UI)
- Soft cartoon backgrounds, readable silhouette

Do not build 3D mesh characters or 3D physics as the main look.

Animation priority

1. Rope snip + 2D split/fall pieces
2. Box step-down tween each mid cut
3. Final heavy drop + land dust (2D particles)
4. Cat 2D states: IdleHang / HopeReact / HappyFree / Sad
5. Scissors 2D sprite anim ~0.2s
6. UI win panel tween

Tools

- Unity **2D URP** project
- **SpriteRenderer** + Sorting Layers (`Background` , `Props` , `Ropes` , `Box` , `Animal` , `FX` , `UI`)
- **Animator** + sprite sheets / Animation Clips for cat/dog/scissors
- DOTween for box Y steps + scissors move + UI
- LineRenderer or 2D segments for ropes
- Unity **2D** particles for snip/land/hearts

2D cat animation set (minimum)

Clip	Feel
`IdleHang`	Soft bob / blink loop
`HopeReact`	Ears up / small bounce on mid cut
`Sad`	Droop for wrong-rope fail (L6+)
`HappyFree`	Jump / celebrate after final land

Same pattern for Dog later (swap sprite set, reuse Animator controller if possible).

Audio

Mid snip variants, stronger final snip, land thud, meow/bark, win jingle, Oops boing.

Art

- Open basket/box as **2D sprites** (front visible)
- Cute cat/dog **sprite sheets**

- Pegs, scissors, FX as 2D
- Commercial-safe free/paid 2D assets; license log
- Optional: Spine/DragonBones later — **not required for v1** (Unity Animator flipbooks enough)

34. Camera, Input, UI, Performance Rules

Orthographic camera framed so full drop path is visible (high box → ground).
No perspective 3D camera for gameplay. Soft 2D background; one finger swipe; big UI; safe area; no star/attempt widgets; limit 2D particles; test real device.

G. DELIVERY

35. Build Order (High Level)

Phase	Focus
1 Fun core	Box + cat + cut + drop steps + final free
2 Content	10 then 20–30 levels, chains, Dog unlock
3 Meta UI	Home, Level Select, Save, Settings, tutorial
4 Juice	Polish snip/drop/cat before ads
5 Ads	Hint rewarded + interstitial rules
6 Auth	Google + cloud merge
7 Polish	Music, settings complete, performance, QA
8 Release	Store assets, privacy, real IDs, internal → production

Do not block Level 1 fun on Firebase/Ads. Guest-first.
Follow §35A step order exactly.
After STEP 28 the game is ready to publish on Google Play.

35A. Detailed Implementation Build Steps (Step 0–28 → Play Store Ready)

> No timeline. Finish each step, test, then next.
> If feel is bad, fix current step — do not add features ahead.
> End state of STEP 28 = Play Store production-ready build.

Global implementation rules

1. Never start with Firebase/Ads
2. Perfect one cut→drop feel before 30 levels
3. Cat always fully visible in open box
4. Test Restart after every new feature
5. No ads on Oops/fail/Restart
6. Put constants in `GameConfig.cs`:
   - `WrongRopeFailFromLevel = 6`
   - `DogUnlockLevel = 10`
   - `InterstitialEveryWins = 4`
   - `InterstitialProtectWins = 8`

Script create order

1. `SwipeCutInput`
2. `RopeController`
3. `ScissorsFX`
4. `HangBoxController`
5. `AnimalController`
6. `LevelManager` / cut-order controller
7. `LevelData` + `LevelLoader`
8. `SaveManager`
9. `GameUI` (Win/Pause/HUD)
10. `AdsManager`
11. `AuthManager` + `CloudSaveService`

STEP 0 — Project Setup

Goal: Clean project ready.

Build

1. Install Unity Hub + Unity LTS

2. Create project: **2D (URP)** — locked (Cut the Rope style; do not use 3D template)
3. Game view portrait 1080×1920, **orthographic** camera
4. Folders: `Art/Sprites`, `Art/Animations`, `Audio`, `Prefabs`, `Scripts` (`Core`, `Cut`, `Rope`, `Box`, `Animal`, `Level`, `UI`, `Save`, `Ads`, `Auth`), `ScriptableObjects/Levels`, `Scenes`
5. Set Sorting Layers: Background, Props, Ropes, Box, Animal, FX, UI
6. Import DOTween free
7. Scenes: `Splash`, `Home`, `Game`
8. Git / GitHub Desktop backup
9. Add empty `GameConfig.cs` with constants above

Done when: 2D URP project opens, sorting layers set, folders/scenes exist, DOTween imported.

STEP 1 — Empty Play Stage (2D)

Goal: Orthographic camera + ground + placeholder 2D box/cat.

Build in `Game` scene

- Orthographic camera framed for full drop path (top → ground)
- 2D background + ground sprites
- Placeholder **SpriteRenderer** open box
- Placeholder **SpriteRenderer** cat as child of box
- Optional 2D guide rails

Animation: none
Done when: Box+cat readable in 2D with space above/below for drops.

STEP 2 — Cat 2D Idle Animation

Goal: Cat feels alive like Cut the Rope characters (2D).

Script: `AnimalController.cs`
Functions: `PlayIdle()`, `PlayHope()`, `PlaySad()`, `PlayHappy()`
Build

- Sprite sheet or simple multi-frame clips
- Unity **Animator** controller with Idle / Hope / Sad / Happy
- Optional tiny DOTween bob on transform (extra life)

Animation: 2D Idle loop first; other clips can be placeholders then polish
Done when: 2D idle loops and looks cute (no 3D model).

STEP 3 — One Static Rope (No Cut)

Goal: Rope draws peg → box.

Build: Peg + `LineRenderer` rope
Functions: `Init(peg, boxAnchor)`, `Draw()`
Animation: optional tiny sway
Done when: Thick readable rope visible.

STEP 4 — Swipe Input Only

Goal: Detect finger/mouse swipe path.

Script: `SwipeCutInput.cs`
Functions: begin / move / end; `GetSwipeSegments()`
Test: debug draw swipe in Editor
Done when: Swipe path reliable.

STEP 5 — Cut Hit Test (No Split)

Goal: Detect swipe vs rope.

Functions: `IsHitBySwipe(...)`, `GetHitPoint()`
Done when: Crossing rope logs HIT; misses do not.

STEP 6 — Scissors + Snip Sound (+ AudioManager foundation)

Goal: Cut juice + audio system started correctly.

Build

- `ScissorsFX` prefab
- `AudioManager` (DontDestroyOnLoad singleton-lite)
  - Separate channels: Music / SFX
  - Read `music_on` / `sound_on` from save (defaults ON)

Functions

- `ScissorsFX.PlayAt(position)`
- `AudioManager.PlaySnip()`
- `AudioManager.PlaySFX(clip)`
- `AudioManager.SetMusicEnabled(bool)`
- `AudioManager.SetSoundEnabled(bool)`
- `AudioManager.PlayMusic(clip, fade)` / `StopMusic(fade)` (can stub music clip later)

Animation: scissors in → close 0.15–0.20s → hide + particles
Done when: Hit feels satisfying; muting SFX via code flag works.

STEP 7 — Rope Split + Fall

Goal: Real cut.

Functions: `CutAt(point)`, `SpawnFallingPieces()`, `FadeAndDestroyPieces()`
Animation: split 2 pieces, DOTween fall+fade (never instant destroy)
Done when: Cut looks like Cut-the-Rope snip/fall.

STEP 8 — Box Step Drop (Mid Cut)

Goal: Non-final cut lowers box one step.

Script: `HangBoxController.cs`
Functions: `SetHeightIndex(i)`, `DropOneStep()`, `PlaySmallTiltThenStraighten()` (optional), `ResetToStart()`
Animation: Y step down 0.25–0.45s + cat `PlayHope()`
Done when: Box clearly lowers and stays readable/upright.

STEP 9 — Two Ropes + Order Logic

Goal: Must cut 1 then 2.

Scripts: `LevelManager` / order controller
Functions: `GetActiveOrder()`, `CanCut(rope)`, `OnRopeCutSuccess(rope)`, `IsFinalRope(rope)`
Wrong rope: Levels 1–5 ignore (use `GameConfig.WrongRopeFailFromLevel`)
Done when: Rope 2 cannot cut before Rope 1.

STEP 10 — Final Cut Full Drop + Free + Win UI

Goal: Complete win loop.

Functions: `FinalDrop()`, `PlayHappy()`, `Win()`
Animation sequence: stronger snip → full drop → land dust → cat happy/jump out → Win panel
Win UI: "Freed!" + Next / Replay / Home (no stars)
Done when: cut → step drop → final drop → free → Next works.

STEP 11 — Restart + Pause

Functions: `RestartLevel()`, `Pause()`, `Resume()`
Behavior: full state restore; no ads
Done when: Restart always clean.

STEP 12 — Soft Fail for Levels 6+

Functions: `OnWrongRope()`, `PlayFail()`, `PlaySad()`, `ResetLevelState()`
Animation: box wobble + sad cat 0.5–1s + Oops + auto reset
Rules: only if `levelId >= 6`; no ad
Done when: Wrong rope fails softly on 6+, ignored on 1–5.

STEP 13 — Level Data System

Goal: Data-driven levels.

Build: `LevelData` SO + `RopeData` + drop heights + `LevelLoader`
Functions: `LoadLevel(id)`, `ClearLevel()`, `SpawnBoxCatRopes()`
Done when: Editing SO changes level without code rewrite.

STEP 14 — First 10 Levels Content

Create `Level_01` … `Level_10`:

- L1–L5: 2 ropes, wrong rope ignore, very easy
- L6–L10: 3 ropes, wrong rope fail enabled
- Clearing 10 sets dog unlock ready

Done when: Can play 1→10 with Next.

STEP 15 — Tutorial Hand (Level 1)

Script: `TutorialHandUI`
Functions: `ShowSwipeOnRope(rope1)`, `HideAfterFirstSuccess()`
Done when: Level 1 understandable with almost no text.

STEP 16 — Splash + Scene Flow + Local Save Core

Goal: App boots cleanly and remembers progress.

Build

- `Splash` scene: logo 1–2s → load Home
- `Bootstrap` / init order: `SaveManager.Load` → `AudioManager.ApplySettings` → go Home
- `SaveManager` local JSON or PlayerPrefs (schema v4)

Functions

- `SaveManager.Load()` / `Save()`
- `SaveLevelCleared(id)`
- `IsLevelUnlocked(id)`
- `IsDogUnlocked()`
- `SaveSettings(music, sound, haptics)`

Done when: Kill app mid-progress → reopen → unlocked/cleared levels correct.

STEP 17 — Home + Level Select UI

Build

- Home: title, big **Play**, **Settings**, optional guest "Save progress" chip placeholder
- Level Select: Cat/Dog sections, checkmarks, locks, "Clear Level 10 to unlock Dog"
- Button punch tweens (DOTween)

Functions

- `OpenLevelSelect()`, `StartLevel(id)`, `OnDogLockedClicked()` message

Done when: Can navigate Splash → Home → Level Select → Game → back without bugs.

STEP 18 — Full Settings (Music / Sound / Haptics / Links)

Goal: Complete settings players expect before store release.

Build Settings screen

- Music toggle (ON/OFF) → stops/starts BGM immediately
- Sound toggle (ON/OFF) → mutes all SFX including snip/UI
- Haptics toggle (ON/OFF) → vibration on cut/fail if device supports
- Privacy Policy button (URL)
- Terms button (URL optional)
- Account section placeholders (wired in Step 23)
- Back button

Functions

- `OnMusicToggled(bool)`

- `OnSoundToggled(bool)`
- `OnHapticsToggled(bool)`
- `OpenPrivacyPolicy()`
- Persist via `SaveManager` on every change

Audio content to add now

- Soft loop BGM for Home + Game (quiet under SFX)
- SFX set: snip×3, final snip, chain clank, land, hope meow, sad, happy, win jingle, Oops, UI click

Done when: Toggles work instantly, survive restart, music ducks under snips.

STEP 19 — Juice Pass (Feel Polish Before Monetization)

Polish:

- Snip variants + final snip stronger
- Box step easing + final heavy drop
- Cat hope/sad/happy
- Land dust + hearts
- UI panel fades / button punches
- Haptics on cut + final drop (if enabled)

Done when: 5 testers say cute / satisfying / one more.

STEP 20 — Expand Content to 20–30 + Chains + Dog Chapter

Build

- Levels 11–30 data
- Chain rope type + clank
- Harder peg layouts
- Dog chapter levels after unlock
- Dog unlock popup "Dog Unlocked!" once

Done when: Full v1 content playable end-to-end as Guest.

STEP 21 — Hint System (Logic First, Ads Hook Second)

Build

- Hint button on HUD
- Glow active rope 3–5s

- Temporary: free hints OR stub "ad success" callback

Functions

- `RequestHint()`
- `HighlightRope(orderIndex)`
- `ClearHighlight()`

Done when: Hint clearly shows next rope; does not auto-cut.

STEP 22 — Ads Integration (Test IDs)

Script: `AdsManager`
Functions

- `InitAds()`
- `ShowHintRewarded(onSuccess, onFail)`
- `TryShowInterstitialOnWin()`
- `IsInterstitialAllowed()`

Rules

- Protect first `InterstitialProtectWins` (8)
- Show every `InterstitialEveryWins` (4) after that
- Never on Oops / Restart / Pause
- Banner not in gameplay
- Use **AdMob test IDs** only in this step

Wire

- Hint button → rewarded
- Win Next path → maybe interstitial then load next

Done when: Test ads show/close cleanly; game resumes correctly; no soft-lock.

STEP 23 — Google Login + Cloud Save

Scripts: `AuthManager`, `CloudSaveService`, `ProgressMerger`, Account UI
Functions: SignInGoogle, SignOut, Pull, PushDebounced, Merge, DeleteAccount
UX

- Soft prompt after Level 5 / Home banner
- Settings: Sign in / Sign out / Delete account
- Never forced before Level 1

Done when: Guest progress merges; new device restore works; delete removes cloud.

STEP 24 — Privacy Consent + Legal Links Live

Build

- First-launch privacy continue gate
- Live Privacy Policy URL + Terms URL
- Data Safety notes prepared (ads, account, cloud progress)
- AdMob UMP / consent flow if required by your AdMob setup

Done when: Fresh install shows consent; links open; account delete path documented.

STEP 25 — Performance, Soft-Lock, and Device QA Pass

Fix checklist

- ☐ 30–60 FPS on target phone during cuts/particles
- ☐ No input during scissors/box tween/win
- ☐ App background/resume safe
- ☐ Call interrupt / ads close resume OK
- ☐ Long session no leak (play 30 levels)
- ☐ Notch/safe area OK
- ☐ Music/SFX toggles still correct after ads
- ☐ Android Back: Gameplay→Pause→Home behavior OK
- ☐ Wrong rope 1–5 ignore / 6+ fail verified
- ☐ Dog unlock once + Dog levels playable

Done when: No critical bugs on your phone + at least one other Android if possible.

STEP 26 — Store Assets + ASO Package

Create

- App icon 512
- Feature graphic 1024×500
- 4–8 gameplay screenshots (real game, hanging box visible)
- Short description + full description (English)
- Optional 15–30s gameplay video (snip → drop steps → free)
- Final application id / package name set in Player Settings
- VersionName + VersionCode
- Portrait only
- Keystore / Play App Signing ready

Done when: All listing assets exported and package id final.

STEP 27 — Release Build + Internal Testing Track

Build

1. Switch AdMob to **production ad unit IDs** (keep a debug flag if needed)
2. Remove debug cheats / verbose logs
3. IL2CPP + ARM64
4. Build AAB (Play requires AAB)
5. Upload Play Console Internal testing
6. Fill Data Safety, content rating, privacy URL, target audience
7. Install from Internal track and smoke-test full flow:
    - New install → tutorial → win → settings music/sound → hint ad → interstitial cadence → Google sign-in → restore

Done when: Internal testers can install and complete core loop without blockers.

STEP 28 — Production Release (Game Ready for Play Store)

Final gate — all must be YES

- ☐ Core loop fun and stable
- ☐ 20–30 levels + Dog chapter
- ☐ Music + Sound + Haptics settings complete and saved
- ☐ Ads fair and working with real IDs
- ☐ Google login optional + cloud merge + delete account
- ☐ Privacy/Terms live
- ☐ Store listing complete
- ☐ Internal test passed
- ☐ VersionCode incremented
- ☐ No known critical crashes

Actions

1. Promote Internal → Closed (optional) or Production
2. Roll out 20% then 100% (safer) or full production
3. Monitor crashes/ANRs first 48 hours
4. Hotfix if needed (new VersionCode)

Done when: App is **Published / In production** on Google Play and installable by public users.

Step completion gates (quick)

After Step	Must feel true
7	Snip cut satisfying
8	Drop step obvious
10	Want to tap Next
14	10 levels not confusing
18	Settings music/sound perfect
19	Friends smile / replay
20	Full content ready
22–23	Ads/login don't hurt fun
25	No critical device bugs
27	Internal track OK
28	**Live on Play Store**

Play Store Ready Definition (end of STEP 28)

Your game is considered fully ready when a new user can:

1. Install from Play Store
2. Open → privacy continue → Home
3. Play Level 1 tutorial → cut → box drops → free cat
4. Use Settings Music/Sound/Haptics
5. Progress saves
6. See fair ads only after protection rules
7. Optionally Sign in with Google and keep progress
8. Reach Dog chapter after Level 10
9. No stars/lives systems
10. No soft-locks after ads or pause

That is the finish line of this build plan.

---

36. Testing & Acceptance

Gameplay

- [ ] Cut/Oops/Win/Next
- [ ] Full cat always visible in open hanging box
- [ ] Mid cuts lower box one clear step
- [ ] Only final cut fully drops box and frees cat
- [ ] Numbered order (only next rope active)
- [ ] Levels 1–5: wrong rope ignored
- [ ] Levels 6+: wrong rope soft fail (sad cat + reset, no ad)
- [ ] Snip + cat reaction juice on each cut
- [ ] No stars UI
- [ ] No attempts UI
- [ ] Dog unlock at 10

Guest save

- [ ] Kill app → progress remains
- [ ] Uninstall warning known (local loss)

Google account

- [ ] Sign in success
- [ ] Cancel sign-in safe
- [ ] Offline sign-in soft fail
- [ ] Merge keeps higher progress
- [ ] New device restore
- [ ] Sign out
- [ ] Delete account removes cloud

Ads

- [ ] Hint rewarded
- [ ] Interstitial only after wins
- [ ] Never on Oops/Restart
- [ ] First ~8 wins protected
- [ ] Ads close → game resumes, no soft-lock

Settings / Audio / Feel

- [ ] Music ON/OFF immediate + persists

- [ ] Sound ON/OFF immediate + persists
- [ ] Haptics ON/OFF + persists
- [ ] Privacy / Terms links open
- [ ] Home + Game BGM loops quietly
- [ ] Snip / win / fail SFX present
- [ ] Android Back behaves correctly

Device / Performance

- [ ] Safe area / notch OK
- [ ] Resume after background OK
- [ ] 30+ FPS during cuts on target phone
- [ ] Long session (20+ levels) stable

Play Store readiness (STEP 28)

- [ ] Production AdMob IDs
- [ ] AAB signed, ARM64
- [ ] Listing assets complete
- [ ] Data Safety + content rating filled
- [ ] Internal track smoke-tested
- [ ] Production published / installable

---

37. Play Store Requirements

- Package name, versioning, signing
- Icon, feature graphic, screenshots
- Privacy Policy URL
- Data Safety: account info, progress cloud, ads data
- Account deletion pathway declared
- Content rating
- Target API requirements
- 64-bit
- Music / Sound / Haptics settings shipped
- Internal testing → Production (see §35A Steps 26–28)

---

38. Marketing Ads Spec

15s Mob Control–style ad (real gameplay)

1. 0–1s: hanging open box + cute cat high up
2. 1–8s: quick ordered snips, box dropping step by step
3. 8–12s: final rope cut, full drop, cat free celebration
4. 12–15s: harder rope/chain flash + logo

Muted-friendly. No menus. Real capture preferred.

---

39. Risks & Edge Cases

Risk	Mitigation
Forced login hurts downloads	Guest-first
Merge wipes progress	max/OR merge policy + tests
Firebase misconfig SHA	Add debug+release SHA
Too many Firestore writes	Debounce pushes
Ad spam churn	Fair cadence
Cut feel weak	Fix before account work
Privacy rejection	Policy + delete account + Data Safety accuracy
“Why didn’t box fall earlier?”	Clear step drops + final-only full drop; optional height marks
Cat hard to see	Always use open basket/box
Boring repetition	Vary peg layouts, chain finale, dog chapter, strong juice

---

40. Later Roadmap

- Spinner
- Remove Ads IAP
- iOS + Apple Sign-In if required
- More animals
- Better rope physics
- Play Games Services achievements optional

- **Do not re-add stars/attempts unless plan explicitly changes**

41. Glossary

Term	Meaning
Open hanging box	Basket/box with visible cat, held by ropes
Drop step	Box lowers one notch after a mid cut
Final cut	Last rope; full drop + free
Active rope	Next numbered rope that can be cut
Wrong rope	Swipe on a rope that is not the active order
Wrong-rope ignore	Levels 1–5: no fail
Wrong-rope fail	Levels 6+: sad cat + reset
Guest	Plays without Google login
Auth uid	Google/Firebase user id
Cloud save	Firestore progress doc
Merge	Combine local+cloud safely
Cleared	Level won ≥ once
Oops	Wrong cut reset, unlimited
Rewarded	Optional ad for Hint

42. One-Page Final Summary

Game: Cut numbered ropes on a hanging open box; each cut lowers it; final cut drops it and frees the cute cat
Fun: Snip + visible drop + cat reaction + big final free (Mob Control–style dopamine; no stars/attempts)
Screen: Pause / Level / Hint / Restart + center box/cat/ropes
Progress: Clear levels; Dog after Level 10
Account: Guest default + optional Google Sign-In
Data: Local always; cloud when signed in; merge keeps best
Ads: Hint + every **4** wins after ~8 protect; never on Oops

Build: Follow §35A Steps 0–28 (cut feel → drop → 10 levels → music/settings → juice → content → ads → auth → QA → Play Store)
Wrong rope: Ignore on 1–5; soft fail from 6+

Requirements Trace (quick)

Need you asked	Covered in
No stars	§3, §11, FR/UI
No attempts	§3, §12
Hanging open box idea	Locked fantasy, §7, §7A
Each cut lowers box	§7, §8, FR-G06
Final cut full drop + free	§7, §8, FR-G07
Mob Control–style interest	§7B, §38
Gaming screen layout	§7A, §24
Google login	§15–16, FR-ACCOUNT
User data local/cloud	§17–18
How game works	§7–14
Screens flow	§22–24
Code structure	§28–30
How to build	§35, **§35A Steps 0–28 → Play Store ready**
Wrong rope rules	§3, §8.3, §12
2D Cut the Rope animation	§3, §10, §26, **§33**, Steps 0–2
Exact asset file names	**§44**
Which function in which script	**§45**

43. Still Pending (Does Not Block Coding)

These items are **not decided yet** or wait until later steps.
You can start Unity coding now (STEP 0) without finishing this list.

Decide while building (non-blocking)

Pending item	When to decide	Notes
Final Play Store name	STEP 26 listing	Working title **Cut Free Pets** is OK until then
Android package name (`com.xxx…`)	STEP 0 draft → lock STEP 26	Can change early; hard after publish
Box motion: small tilt→straighten **vs** rails upright	STEP 8–9 while tuning feel	Either is fine; pick what looks clearer
Exact Level 1–10 / 11–30 peg layouts	STEP 14 + STEP 20	Design on paper then fill `LevelData`
Privacy Policy / Terms **live URLs**	STEP 24–27	Placeholder link OK until release
Google Play Console account (**$25** one-time)	Before STEP 27 upload	Not needed for local Unity work
Real AdMob app + ad unit IDs	STEP 27 release build	Use **test IDs** in STEP 22
Firebase project + SHA keys for Google Sign-In	STEP 23	After core fun works
App icon / screenshots / feature graphic	STEP 26	Capture from real gameplay
Keystore / Play App Signing	STEP 26–27	Before AAB upload
Soft privacy consent copy (final wording)	STEP 24	Keep short and honest

Already locked (do not reopen without updating this file)

- Hanging open box + numbered cuts + drop steps + final free
- **2D** Cut the Rope–style animation (Unity 2D URP)
- No stars / no attempts
- Wrong rope: ignore L1–5 / soft fail L6+
- Dog unlock at Level 10
- Guest-first + optional Google + cloud merge
- Ads: hint rewarded + interstitial after protect wins
- Full build path Steps **0–28** → Play Store ready

Start coding checklist

- [ ] Unity Hub + Unity LTS installed
- [ ] Create **2D (URP)** project
- [ ] Open this file → follow **§35A STEP 0**

When a pending item is decided, write the decision into §3 Locked Decisions and remove/update it here.

---

44. Asset File Catalog (Exact Names)

> Use these **exact file names** so art, audio, and code stay consistent.
> Placeholder colored rectangles are OK until final art — keep the names the same.
> All visuals are **2D** (sprites / sprite sheets), Cut the Rope style.

44.1 Folder layout

```
Assets/
  Art/
    Sprites/
      Characters/
      Props/
      FX/
      UI/
      Backgrounds/
    Animations/          # .controller + Animation clips
    Backgrounds/         # optional if not under Sprites
  Audio/
    Music/
    SFX/
  Prefabs/
  ScriptableObjects/Levels/
  Scripts/               # see §45
```

44.2 Character sprites (Cat)

Purpose	File name(s)	Notes
Idle frames	`cat_idle_01.png` … `cat_idle_04.png`	Soft blink/bob loop
Hope / mid-cut react	`cat_hope_01.png` … `cat_hope_03.png`	Ears up / small bounce
Sad / fail	`cat_sad_01.png` … `cat_sad_03.png`	Levels 6+ wrong rope
Happy / free	`cat_happy_01.png` … `cat_happy_06.png`	Final land celebration
Optional sheet	`cat_sheet.png`	One spritesheet instead of separate frames OK

44.3 Character sprites (Dog — after unlock)

Purpose	File name(s)
Idle	`dog_idle_01.png` … `dog_idle_04.png`
Hope	`dog_hope_01.png` … `dog_hope_03.png`
Sad	`dog_sad_01.png` … `dog_sad_03.png`
Happy	`dog_happy_01.png` … `dog_happy_06.png`
Optional sheet	`dog_sheet.png`

44.4 Props / gameplay sprites

Purpose	File name
Open hanging box / basket	`box_open.png`
Box shadow (optional)	`box_shadow.png`
Peg / anchor	`peg.png`
Rope segment / texture	`rope_segment.png`
Chain segment	`chain_segment.png`
Rope number badge 1–5	`num_1.png` … `num_5.png`
Scissors snip frames	`scissors_01.png` … `scissors_04.png`
Tutorial hand	`hand_tutorial.png`

44.5 FX sprites

Purpose	File name
Snip spark	`fx_snip_spark.png`
Land dust	`fx_land_dust.png`
Hearts	`fx_heart.png`
Rope glow (hint)	`fx_rope_glow.png`

44.6 UI sprites

Purpose	File name
Play button	`btn_play.png`
Pause	`btn_pause.png`

Hint	`btn_hint.png`
Restart	`btn_restart.png`
Settings	`btn_settings.png`
Next	`btn_next.png`
Lock	`icon_lock.png`
Cleared check	`icon_check.png`
Music on/off	`icon_music_on.png` , `icon_music_off.png`
Sound on/off	`icon_sound_on.png` , `icon_sound_off.png`
Haptics on/off	`icon_haptics_on.png` , `icon_haptics_off.png`
App icon source	`app_icon_1024.png` (export 512 for Play)

44.7 Backgrounds

Purpose	File name
Splash / logo	`bg_splash.png` , `logo_cutfreepets.png`
Home	`bg_home.png`
Gameplay Cat chapter	`bg_game_cat.png`
Gameplay Dog chapter	`bg_game_dog.png`
Ground / floor	`ground.png`

44.8 Unity Animation assets (2D Animator)

Asset name	Path / type	Clips inside
`Anim_Cat.controller`	`Art/Animations/`	`Cat_Idle` , `Cat_Hope` , `Cat_Sad` , `Cat_Happy`
`Anim_Dog.controller`	`Art/Animations/`	`Dog_Idle` , `Dog_Hope` , `Dog_Sad` , `Dog_Happy`
`Anim_Scissors.controller`	`Art/Animations/`	`Scissors_Snip`
`Cat_Idle.anim`	clip	uses `cat_idle_*`
`Cat_Hope.anim`	clip	uses `cat_hope_*`
`Cat_Sad.anim`	clip	uses `cat_sad_*`
`Cat_Happy.anim`	clip	uses `cat_happy_*`
`Scissors_Snip.anim`	clip	uses `scissors_01` … `04` , ~0.2s

Animator parameter suggestion: string/trigger `State` or triggers `Hope`, `Sad`, `Happy` → return to Idle.

44.9 Audio files

Purpose	File name
Home BGM	`bgm_home.ogg`
Game BGM	`bgm_game.ogg`
Mid snip variants	`sfx_snip_01.ogg`, `sfx_snip_02.ogg`, `sfx_snip_03.ogg`
Final snip	`sfx_snip_final.ogg`
Chain clank	`sfx_chain_clank.ogg`
Box land	`sfx_land.ogg`
Cat hope	`sfx_meow_hope.ogg`
Cat sad	`sfx_meow_sad.ogg`
Cat happy	`sfx_meow_happy.ogg`
Dog bark happy (later)	`sfx_bark_happy.ogg`
Win jingle	`sfx_win.ogg`
Oops	`sfx_oops.ogg`
UI click	`sfx_ui_click.ogg`

44.10 Prefabs (exact names)

Prefab	Contains
`PF_OpenHangBox.prefab`	box sprite + attach points + `HangBoxController`
`PF_Cat.prefab`	SpriteRenderer + Animator(`Anim_Cat`) + `AnimalController`
`PF_Dog.prefab`	same for dog
`PF_Rope.prefab`	LineRenderer/segments + `RopeView` + number badge
`PF_Chain.prefab`	chain variant of rope
`PF_Peg.prefab`	peg sprite
`PF_ScissorsFX.prefab`	scissors sprites + Animator + `ScissorsFX`
`PF_SnipParticles.prefab`	cut FX
`PF_LandDust.prefab`	land FX

`PF_TutorialHand.prefab`	hand + tween

44.11 Level ScriptableObjects

Asset	Name pattern
Levels 1–10	`Level_01.asset` … `Level_10.asset`
Levels 11–30	`Level_11.asset` … `Level_30.asset`
Dog chapter	`Level_D01.asset` … (or continue numeric after unlock gate)

44.12 Minimum v1 art pack (make or buy first)

1. `box_open.png` + `peg.png`
2. Cat idle + hope + sad + happy (even 2 frames each is OK early)
3. `scissors_01` … `04`
4. Simple `bg_game_cat.png` + `ground.png`
5. Snip + land + win SFX (placeholders OK)

---

45. Script → Function Map (Which Function in Which File)

> Put each function in the listed `.cs` file. Do not dump everything into one God script.

45.1 Core / config

File	Path	Functions / contents
`GameConfig.cs`	`Scripts/Core/`	Constants only: `WrongRopeFailFromLevel=6`, `DogUnlockLevel=10`, `InterstitialEveryWins=4`, `InterstitialProtectWins=8`
`Bootstrap.cs`	`Scripts/Core/`	`Awake` init order: Save → Audio.ApplySettings → Auth.Init → Ads.Init → load Home
`GameSession.cs`	`Scripts/Core/`	`StartLevel(id)`, `RestartLevel()`, `Pause()`, `Resume()`, `Win()`, `FailOops()`, `ReturnHome()`

45.2 Cut / input / rope

File	Path	Functions
`SwipeCutInput.cs`	`Scripts/Cut/`	`OnPointerDown/Drag/Up` (or Update touch), `GetSwipeSegments()`, `IsSwipeActive()`

RopeView.cs	Scripts/Rope/	Init(peg, boxAnchor, orderIndex, type), Draw(), IsHitBySwipe(segments), GetHitPoint(), CutAt(point), SpawnFallingPieces(), FadeAndDestroyPieces(), SetActiveVisual(bool), SetHintGlow(bool)
CutOrderController.cs	Scripts/Cut/	GetActiveOrder(), CanCut(rope), OnRopeCutSuccess(rope), IsFinalRope(rope), OnWrongRope(), ResetOrder()
ScissorsFX.cs	Scripts/Cut/	PlayAt(Vector3 position)

45.3 Box / animal

File	Path	Functions
HangBoxController.cs	Scripts/Box/	SetHeightIndex(i), DropOneStep(), FinalDrop(), PlaySmallTiltThenStraighten(), PlayFailWobble(), ResetToStart(), GetAttachPoint(index)
AnimalController.cs	Scripts/Animal/	PlayIdle(), PlayHope(), PlaySad(), PlayHappy(), SetAnimalType(Cat/Dog)

45.4 Level

File	Path	Functions
LevelData.cs	Scripts/Level/	ScriptableObject fields only (levelId, ropes, dropStepHeights, …)
RopeData.cs	Scripts/Level/	Serializable rope fields
LevelLoader.cs	Scripts/Level/	LoadLevel(id), ClearLevel(), SpawnBoxCatRopes(), GetCurrentLevelData()

45.5 UI

File	Path	Functions
GameHUD.cs	Scripts/UI/	SetLevelLabel, OnPauseClicked, OnHintClicked, OnRestartClicked
PauseUI.cs	Scripts/UI/	Show(), Hide(), OnResume, OnHome, OnSettings
WinUI.cs	Scripts/UI/	Show(), OnNextClicked
OopsUI.cs	Scripts/UI/	ShowBrief() (toast/overlay; no ad)
HomeUI.cs	Scripts/UI/	OnPlay, OnSettings, RefreshAccountChip
LevelSelectUI.cs	Scripts/UI/	RefreshGrid(), OnLevelClicked(id), OnDogLockedClicked()

SettingsUI.cs	Scripts/UI/	OnMusicToggled, OnSoundToggled, OnHapticsToggled, OpenPrivacyPolicy, OpenTerms
TutorialHandUI.cs	Scripts/UI/	ShowSwipeOnRope(rope), HideAfterFirstSuccess()
AccountUI.cs	Scripts/UI/	OnSignIn, OnSignOut, OnDeleteAccount
PrivacyGateUI.cs	Scripts/UI/	ShowIfNeeded(), OnContinue

45.6 Audio

File	Path	Functions
AudioManager.cs	Scripts/Audio/	PlaySnip(), PlaySnipFinal(), PlaySFX(clipOrKey), PlayMusic(clip, fade), StopMusic(fade), SetMusicEnabled(bool), SetSoundEnabled(bool), ApplySettings(), PlayUIClick()

45.7 Save

File	Path	Functions
SaveManager.cs	Scripts/Save/	Load(), Save(), SaveLevelCleared(id), IsLevelUnlocked(id), IsDogUnlocked(), SaveSettings(music, sound, haptics), GetProfile()
LocalSaveRepository.cs	Scripts/Save/	ReadJson(), WriteJson()

45.8 Ads

File	Path	Functions
AdsManager.cs	Scripts/Ads/	InitAds(), ShowHintRewarded(onSuccess, onFail), TryShowInterstitialOnWin(), IsInterstitialAllowed()

45.9 Auth / cloud

File	Path	Functions
AuthManager.cs	Scripts/Auth/	Init(), SignInGoogle(), SignOut(), IsSignedIn, GetUserId()
CloudSaveService.cs	Scripts/Cloud/	Pull(), Push(), PushDebounced(), DeleteCloudAccountData()
ProgressMerger.cs	Scripts/Cloud/	Merge(local, cloud) → merged profile
NetworkStatus.cs	Scripts/Core/	IsOnline() soft check

45.10 Optional

File	Path	Functions
`AnalyticsManager.cs`	`Scripts/Analytics/`	`LogEvent(name, params)`
`HapticsService.cs`	`Scripts/Core/`	`VibrateLight()`, `VibrateMedium()` (respect settings)

45.11 Who calls whom (quick)

```
SwipeCutInput → RopeView.IsHitBySwipe
              → CutOrderController.CanCut
              → if ok: ScissorsFX.PlayAt + AudioManager.PlaySnip + RopeView.CutAt
                       CutOrderController.OnRopeCutSuccess
                       → HangBoxController.DropOneStep OR FinalDrop
                       → AnimalController.PlayHope / PlayHappy
                       → GameSession.Win (if final)
              → if wrong L6+: CutOrderController.OnWrongRope
                       → HangBoxController.PlayFailWobble + AnimalController.PlaySad
                       → GameSession.FailOops / ResetLevelState

WinUI.OnNext → AdsManager.TryShowInterstitialOnWin → LevelLoader.LoadLevel(next)
Hint button  → AdsManager.ShowHintRewarded → RopeView.SetHintGlow on active rope
Settings     → AudioManager.SetMusic/Sound + SaveManager.SaveSettings
Sign-in      → AuthManager → CloudSaveService.Pull → ProgressMerger.Merge → SaveManager.Save
```

45.12 Naming rules

1. One public class per file; file name = class name
2. No stars/attempts fields anywhere
3. Prefab names start with `PF_`
4. Animation controllers start with `Anim_`
5. Audio: `bgm_` music, `sfx_` effects
6. Sprites: `cat_`, `dog_`, `btn_`, `icon_`, `fx_`, `bg_` prefixes

Final Line

Ship **fun first** as Guest:
cut → box drops a step → final cut drops fully → cat free.
Make every cut feel good (like Mob Control dopamine).
Google login protects progress — it never blocks play.
Keep this file updated when any requirement changes.
Use **§44** for every asset name and **§45** for every script function location.

End of Complete Project Requirements & Game Bible v4.4